

- 29** A stationary uranium nucleus of mass 238 units disintegrates by the emission of an α -particle of mass 4 units.

The ratio $\frac{\text{kinetic energy of the } \alpha \text{ - particle}}{\text{kinetic energy of the recoiling daughter nucleus}}$ is

A $\frac{4}{234}$

B $\frac{4}{238}$

C $\frac{234}{4}$

D $\frac{238}{4}$

- 30** The half-life of a certain radioactive material is 3.0 s